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AI-Driven Redesign of Learning Environments: A Cognitive Infocommunications Framework for Structural Education Reform

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Abstract. Contemporary education systems retain an industrial-era architecture that is poorly matched to what cognitive science indicates about individual learning, relying on uniform-pace instruction, age-based grouping, and delayed, opaque assessment. This paper proposes a cognitive infocommunications framework for structural education reform, integrating adaptive AI-driven learning platforms, longitudinal digital portfolio systems, and mixed-age peer apprenticeship models grounded in established cognitive science. Drawing on Bloom's mastery learning paradigm, Vygotsky's zone of proximal development, and the protege effect in peer-mediated learning, the framework redefines the learning environment as an adaptive cognitive infocommunication system. The Israeli education system serves as an illustrative implementation scenario. A phased deployment sequence is outlined together with proposed evaluation metrics grounded in cognitive performance indicators. The paper presents a conceptual architecture; no implementation or empirical validation is claimed.

Keywords: adaptive learning systems, cognitive infocommunications, AI-assisted education, portfolio assessment, mastery learning, peer apprenticeship, education reform

1 Introduction

The architecture of contemporary mass education emerged in response to industrial-era organisational demands rather than from cognitive-developmental principles. The model of centrally prescribed, uniform-pace instruction took shape in the nineteenth century alongside the labour requirements of industrial capitalism [1]. Its design priorities were uniformity and scheduling efficiency. These differ from the adaptive, feedback-rich conditions that cognitive science associates with genuine knowledge acquisition [2].

The cognitive infocommunications (CogInfoCom) framework, established by Baranyi, Csapo, and Sallai as the systematic study of how cognitive processes co-evolve with infocommunication devices [3], provides the theoretical home for this proposal. In CogInfoCom terms, the learning environment is a cognitive infocommunication system in which the capabilities of natural cognitive agents are extended and enhanced through artificial cognitive architectures [3]. More recent statements of the field extend it toward Digital Reality and Cognitive Entities, the latter denoting the synergetic

combination of humans, devices, and environment [4]. The present proposal takes that framework from the level of individual tools to the structure of a national education system.

Israel offers an illustrative case: consistent above-OECD education expenditure reaching 7.67% of GDP at peak has coincided with a decline from top-tier standing in science and mathematics in the 1960s to thirty-third out of forty-one nations by 2002 [5]. The 2005 Dovrat Commission attributed this outcome to structural rather than financial factors [6].

The contribution of this paper is the integration of adaptive AI, peer interaction, competency tracking, and longitudinal assessment into a single CogInfoCom educational architecture. The framework combines four proposed structural components: competency-based rather than age-based grouping; mixed-age peer apprenticeship drawing on the protege effect [7] and Vygotsky's zone of proximal development [8]; an AI-assisted adaptive platform applying Bloom's mastery principle [1][2]; and a longitudinal cumulative portfolio in place of single-point examination verdicts [9]. Each component is presented as a design proposal requiring formal specification and empirical evaluation. The components are ordered in delivery. Learning proceeds on the adaptive platform; a learner who fails to advance there is referred to a near-peer mentor, and expert remediation by the teacher-mentor follows only where peer apprenticeship also fails (Section 5).

2 Related Work

The CogInfoCom field, as defined by Baranyi, Csapo, and Sallai [3], investigates the synergic combination of infocommunication technologies and cognitive sciences to enable cognitive processes to co-evolve with ICT devices. The adaptive learning architecture proposed here applies CogInfoCom principles directly to institutional education design. Within the field, the concept of mathability addresses the emulation and enhancement of human mathematical capability through artificial cognitive systems [10], and provides a domain-specific precedent for the adaptive engine proposed in Section 4. Current work in the field has moved toward treating designed environments themselves as cognitive systems: the programme of Cognitive Aspects of Virtual Reality examines how spatial and technological environments shape cognitive performance [11], while recent field statements foreground human-technology co-evolution and the emergence of cognitive entities spanning human, device, and environment [4]. This paper treats the learning environment as an instance of that broader class, not as an isolated educational technology.

In VanLehn's review of controlled comparisons, intelligent tutoring systems (ITS) with step-based feedback approached the effectiveness of human tutors, at $d = 0.76$ against $d = 0.79$ relative to no tutoring [12]. Neither figure approaches the two-sigma advantage Bloom reported for one-to-one tutoring [2]. Comprehensive treatments of intelligent tutoring architecture set out the design space of such systems, spanning learner modelling, pedagogical strategy, and interface [13]. A later meta-analysis of fifty controlled evaluations reported a median effect of 0.66 standard deviations for intelligent tutoring over conventional instruction, and noted that estimates depend

heavily on whether outcomes are measured with locally developed or standardised instruments [14].

The subsequent emergence of large language models has widened this design space considerably and has raised pedagogical, equity, and governance questions that adaptive systems of the earlier generation did not confront [15]. The architecture proposed here is model-agnostic and does not presuppose any particular generation of underlying technology. Early randomised evidence on such systems points in both directions. In an undergraduate physics course, an AI tutor designed on the same pedagogical principles as the in-class lessons produced greater learning in less time than in-class active learning [16]. In a field experiment with nearly a thousand high school mathematics students, unrestricted access to GPT-4 raised grades on assisted practice problems by 48 percent relative to controls, but lowered grades on a later unassisted exam by 17 percent [17]. In the same trial, a tutor with prompts designed to safeguard learning largely removed that loss. The second study held the model constant and varied only the design, so the difference is attributable to design; this is the premise on which the architecture is treated as model-agnostic.

Corbett and Anderson's knowledge tracing model established the foundational probabilistic framework for modelling student competency states over time [18]. It supplies the theoretical basis for the continuous benchmarking architecture proposed here.

Portfolio-based assessment systems have been explored in several national contexts. Darling-Hammond and Adamson showed that longitudinal performance documentation produces more valid competency inferences than point-in-time standardised testing [9]. Roscoe and Chi's distinction between knowledge-building and knowledge-telling in peer tutoring established the mechanism through which the protege effect produces genuine cognitive consolidation [19]. The existing literature has not integrated these components into a unified system-level architectural framework with defined data models, deployment phases, and evaluation metrics.

3 Group Composition, Peer Tutoring, and Learning Dynamics

That the performance of a group is a measurable property not reducible to the average ability of its members is an established empirical finding. Woolley et al. identified a general collective intelligence factor accounting for group performance across diverse tasks, only weakly predicted by the individual intelligence of members [20]. An early and much cruder formulation of the same intuition appears in Le Bon, who argued that conformity and social contagion can depress collective reasoning below the level its stronger members would reach independently [21]. Le Bon is cited here as a historical antecedent of the group-composition question, not as evidence bearing on educational practice.

Grouping arrangements considered on their own, however, show little effect. Veenman's best-evidence synthesis reported a median effect size of zero for multigrade against single-grade classes and minus 0.03 for multi-age against single-age grouping [22], and meta-analytic work on ability grouping found only small average effects concentrated in enriched provision for high attainers [23]. The robust finding lies elsewhere. A recent meta-analysis of thirty-two cross-age tutoring studies reported a

positive effect on academic outcomes for both parties, with an overall effect size of 0.34, tutees gaining 0.33 and tutors 0.39 [24]. That tutors gain at least as much as tutees is consistent with the protege effect [7] and is the finding on which the present framework principally relies. An updated meta-analysis focused specifically on tutor achievement reported a weighted mean effect size of 0.43 and identified structured tutoring and same-age non-reciprocal tutoring among the parameters associated with stronger outcomes [25]. The same-age finding indicates that the active ingredient is a stable competency asymmetry between the two parties rather than chronological distance as such.

This distinction determines the role of grouping within the proposed framework. Competency-based grouping is not advanced here as beneficial in its own right. It is the condition under which structured peer tutoring can operate as the default mode of a learning environment. Age-stratified organisation does not make cross-age tutoring impossible, as buddy schemes and pull-out programmes demonstrate, but it confines such arrangements to the periphery of the timetable and makes stable competency asymmetry a matter of accident. The asymmetry itself is not scarce within age cohorts. In TIMSS 2019 data, about 37 percent of the variance in US grade 8 mathematics achievement lay within classrooms, and 35 percent of grade 8 classrooms contained students at all four international benchmarks [26]. What age-stratified organisation lacks is a means of identifying and using that asymmetry. Mixed-age composition, on this account, is a consequence of grouping by competency rather than its purpose.

The asymmetry condition is bounded on both sides. A mentor must have mastered the unit the learner is currently attempting, but a mentor too far ahead is less suitable, since the distance between the two representations of the material grows [8]. Stable competency asymmetry therefore means asymmetry within a near-peer band, not any positive gap. In cognitive infocommunications terms [3], the requirement is a system capable of identifying, at any moment and in each domain, which pairings fall within that band. This would be the output of the continuous benchmarking subsystem described in Section 4: a competency estimate maintained per learner and per domain, from which admissible mentor-learner pairings can be derived without reference to birth year. Whether such pairings reproduce the effect sizes observed in the tutoring literature is an empirical question, and one of the principal hypotheses this architecture generates.

4 System Architecture: The AI-Assisted Platform

The proposed cognitive infocommunications architecture comprises three integrated subsystems: an adaptive content delivery engine, a continuous competency benchmarking system, and a transparent assessment and feedback mechanism. In the proposed design, every learner-system interaction would generate data that updates the competency model, adjusts the content pathway, and contributes to the longitudinal portfolio record. None of these subsystems has been implemented. They are specified here at the architectural level. These subsystems operate together within the platform layer; the escalation from platform to peer to teacher-mentor is described in Section 5.

4.1 Adaptive Content Delivery Engine

The proposed content delivery engine would generate problems and scenarios calibrated to the learner's current competency state across four core knowledge domains: language arts, mathematics and geometry, civics, and natural science. The engine would apply a mastery-gated progression model derived from Bloom's paradigm [1][2], in which advancement is conditional on demonstrated competency rather than elapsed time. The competency state is modelled as a dynamic Bayesian network in the tradition of Corbett and Anderson's knowledge tracing framework [18]. It would be updated with each learner interaction. Content difficulty would target the learner's zone of proximal development, operationalising Vygotsky's developmental theory [8] within a computable architecture. In the mathematics domain, this design intent aligns with the mathability agenda of extending human mathematical capability through artificial cognitive support rather than replacing it [10].

4.2 Continuous Competency Benchmarking

The benchmarking subsystem would maintain a running longitudinal record of each learner's competency state, updated with every platform interaction. This trajectory record is intended to drive the content engine's adaptation and to feed the longitudinal portfolio data structure. It would also constitute a data infrastructure for system-level evaluation at a granularity that existing national education systems do not provide [5] [6].

4.3 Assessment and Feedback Mechanism

The assessment subsystem is designed to apply Black and Wiliam's findings that immediate formative feedback is associated with stronger learning outcomes than delayed summative assessment [27], at low marginal cost per additional learner interaction. Three student-initiated milestone assessments are proposed. They are the SSAT from age six to fourteen, the SAT from age fourteen, and the AP from age seventeen, each available in Hebrew, Arabic, or English. The language of examination would be recorded in the portfolio as an additional, verifiable indicator of bilingual academic competency.

5 Peer Apprenticeship as Cognitive Transfer Mechanism

The mixed-age peer apprenticeship model treats the protege effect as a structural component of the learning environment. Bargh and Schul demonstrated that learners who acquire material with the intention of teaching it encode it more deeply and retain it more durably [28]. Chase et al. formalised this as the protege effect: teaching another person deepens the teacher's own understanding and identifies gaps that self-directed study would not reveal [7]. Meta-analytic evidence supports the effect at scale, with tutors gaining at least as much as tutees across cross-age tutoring studies [24][25].

In the proposed architecture, the teacher-mentor would introduce each unit to a learning group, after which learners would work through it on the adaptive platform. Support escalates only on failure. A learner whose competency fails to advance on the platform would be referred to a near-peer mentor, drawn from learners the benchmarking subsystem records as having mastered the unit and as lying within the

admissible band set out in Section 3. Expert remediation by the teacher-mentor would follow only where peer apprenticeship also fails. Mentoring is therefore occasional, triggered by a failure to advance, and pairings would be recomputed from current competency estimates whenever the peer layer is invoked. The teacher-mentor would also supervise the quality of peer interaction. The two expert roles are distinct: framing at unit entry precedes difficulty, whereas remediation responds to it, and only remediation stands at the end of the escalation sequence.

In CogInfoCom terms, this arrangement is a case of inter-cognitive communication, in which two natural cognitive entities with differing internal representations exchange information through an artificially mediated channel [29]. The pairing is selected by the benchmarking subsystem and the platform supplies the shared task representation over which the exchange occurs, so the mediation is constitutive of the interaction rather than incidental to it. The peer apprenticeship model can be described as a distributed cognitive processing network in which knowledge representation is strengthened through the act of transmission [3]. Near-peer mentoring is not proposed as a substitute for expert instruction, which produces the largest effects among tutoring programmes [30]. Its role is to reserve expert time for learners the peer layer cannot serve. Within that role, a mentor's recent acquisition of the material would keep the representation distance to be bridged small [8], which is the reason admissibility is bounded from above. Whether peer mentoring within that band produces advancement is a hypothesis for pilot testing.

6 Longitudinal Portfolio as Data Infrastructure

The portfolio performs what CogInfoCom describes as representation bridging [29]. Competency states held internally by a learner, and estimated probabilistically by the platform, are rendered in a form that institutions with entirely different representational conventions can interpret, including employers, universities, and public bodies. The cumulative longitudinal portfolio is the proposed data architecture integrating the outputs of the three platform subsystems into a unified competency record. It would be a continuously updated digital document, initiated at the beginning of the mandatory learning phase and maintained throughout the learner's educational trajectory.

The portfolio data model comprises five record types: assessment records capturing AI-generated competency evaluations timestamped chronologically; examination records capturing milestone attempts, scores, and languages; apprenticeship records capturing peer mentoring assignments and vocational placement outcomes; enrichment records capturing voluntary learning activities; and observer records capturing teacher-mentor qualitative assessments, flagged as human-generated and distinguished from AI-generated entries.

The portfolio would be owned by the learner and would require an extended privacy legislation framework with access tiers defined by statute. One potential application concerns military intake, where the current three-day psychometric process captures little of a recruit's developmental trajectory. A longitudinal portfolio spanning six to twelve years is hypothesised to improve placement precision by supplementing existing intake instruments. Structured portfolio access for talent identification programmes, subject to parental consent, is likewise proposed as a possible route to broadening

access beyond informal referral networks. Both applications raise substantial governance questions, addressed in Section 8.

7 Implementation Scenario: Israel as Illustrative Case

The following deployment sequence is presented as an indicative scenario intended to expose the institutional and legal questions the architecture raises, not as a validated feasibility assessment. Israel offers a useful illustrative context. Its technology sector could supply AI development capacity and digital infrastructure through public-private partnership, its compulsory military service creates a natural portfolio integration point, and its linguistic plurality tests the multilingual requirements of the proposed platform.

7.1 Indicative Deployment Sequence

A first phase would establish the legislative framework governing portfolio legal standing and privacy architecture, initiate platform procurement, and deploy the system in a small number of pilot municipalities selected for demographic, geographic, and socioeconomic diversity. A second phase would extend the framework more widely and would require a decision on the status of the existing Bagrut examination system. That change has legal and institutional implications well beyond the scope of this paper. A third phase would address portfolio integration with military intake and any associated economic incentive structures. None of this is costed here. Each phase would need its own timeline, and the platform its own cost estimate. The legislative pathway requires legal analysis that this paper does not provide.

7.2 Cost Structure and Fiscal Mechanisms

A structural argument for the fiscal position of the proposed architecture can be stated even though a costed projection cannot. The baseline is unusual: Israel has sustained education expenditure above the OECD average as a proportion of GDP throughout the period for which international data exists, peaking at 7.67% against a world average near 4.15%. Per-student outcomes have nonetheless remained below the OECD average [5][6]. More spending is therefore not the issue. The question is whether the proposed architecture changes the cost structure itself.

Two distinct ledgers must be separated here, since they are borne by different parties. Public expenditure comprises teacher salaries, tertiary subsidy, and remedial and dropout provision, all of which recur annually without producing structural change. Household expenditure comprises the private tutoring economy that families fund directly to compensate for institutional shortfall. The latter does not appear in state accounts at all, with two consequences. Reported public expenditure understates the true cost of operating the system, and the shortfall is met only by those households able to pay for it. The incumbent arrangement thus converts an institutional deficiency into a distributional one. The proposed architecture would exhibit a different profile: capital investment concentrated in the first years, principally platform procurement, portfolio infrastructure, and screening and training of staff, followed by recurring effects that depend on mechanisms rather than on expenditure levels.

Six such mechanisms can be identified. First, redistributing the delivery function across the adaptive platform, peer apprenticeship, and community placement would change the required size of the teaching workforce, which bears on aggregate salary

expenditure even where per-practitioner compensation rises. The change is confined to remediation. Expert framing of new units is unaffected by the peer layer, so any reduction concerns the part of teacher time that scales with individual learner difficulty. Second, platform-delivered practice, assessment, and benchmarking carry low marginal cost per additional learner, whereas the corresponding functions in the incumbent system (preparation, marking, and examination administration) scale with the size of the cohort. Third, accredited community placement transfers part of the delivery cost to the private sector, offset by incentives whose magnitude relative to institutional per-learner cost determines whether the transfer yields a net saving. Fourth, earlier entry to productive work alters the ratio of contributors to dependents in a cohort. Fifth, an honestly differentiated vocational pathway changes the composition of tertiary demand and therefore the distribution of subsidy. Sixth, portfolio-informed military placement bears on training duration and reallocation costs. The first, second, fifth, and sixth mechanisms act on public expenditure directly. The third redistributes cost between the state and private employers, with the incentive magnitude determining the net position, and the fourth acts on public revenue rather than expenditure. A seventh effect falls outside the state ledger entirely. To the extent that continuous formative feedback reduces household reliance on private tutoring, the gain accrues to families and is properly counted as a distributional outcome rather than a fiscal saving.

Each mechanism is a hypothesis with unestimated parameters, and the direction of the aggregate effect cannot be settled by argument. What the architecture does supply is the instrumentation required to settle it: the cost-per-mastered-competency metric defined below is computable from portfolio data and is not computable within an assessment regime that records examination outcomes alone. The fiscal claim advanced here is accordingly narrow. The architecture is designed so that its own cost-effectiveness becomes measurable, and pilot deployment is the means by which the mechanisms above would be tested rather than asserted.

7.3 Proposed Evaluation Metrics and Study Design

Evaluation would be grounded in cognitive performance indicators rather than examination pass rates. Each metric below is stated with its proposed operational definition; all require formal specification and instrument validation before use.

Competency trajectory gradient. The slope of estimated mastery probability against cumulative engaged task time, fitted separately per knowledge domain over rolling windows. The unit is change in mastery probability per hour of active interaction. Interpretation requires comparison against cohort-matched baselines, since raw gradients vary with domain difficulty and entry level.

Competency depth index. The proportion of items first recorded as mastered that remain correct at spaced retention probes administered at thirty, ninety, and one hundred and eighty days. This separates durable acquisition from short-term performance and provides a check against overfitting of the mastery estimate to recent practice.

Peer apprenticeship quality. A difference-in-differences estimate comparing the mentee trajectory gradient during active mentoring assignments against the same learner outside such assignments, controlling for prior gradient. The mentor trajectory gradient in the mentored domain, measured over the same interval, yields a separate estimate of the protege effect [7].

Portfolio richness. A normalised measure of distribution across record types and domain coverage rather than a raw entry count, so that volume of activity alone cannot inflate the indicator.

Cost per mastered competency. Total system cost, including platform development amortisation and staffing, divided by the number of retention-confirmed mastered competency units, enabling comparison against per-learner expenditure in the incumbent system.

Reactivity and measurement contamination. Because the record is intended to carry consequences, it is subject to the distortion Campbell identified in consequential social indicators, whereby an indicator used for decision-making corrupts the process it monitors [31]. In educational assessment this appears as score inflation: gains on the consequential measure that do not transfer to an equivalent measure without stakes [32]. The architecture is exposed to this in two ways. Learners aware that the record is read at intake may alter how they engage with the platform, and teacher-mentors aware of the same may alter what they record. Three design responses follow. A low-stakes audit instrument, administered to a random subsample under controlled conditions and never entering the portfolio, permits drift between the two estimates to be measured directly rather than assumed absent. Rotating item pools reduce the efficiency of coaching to specific items. Machine-scored entries are held separately from human-authored observer entries, the latter being the more inflation-prone component, which supplies a second reason for the exclusion of childhood observer entries from intake access proposed in Section 8.

Study design. Testing these indicators would require a cluster-randomised pilot at school or municipality level with pre-registered hypotheses and matched control sites. Novelty effects, selection into pilot sites, and teacher-level variation are the principal confounds and would require explicit control. Because deployment has not yet occurred, the disclosure regime itself can be randomised across pilot sites, so that cohorts differing in whether the intake use of the record is made salient yield a direct estimate of reactivity rather than a post hoc conjecture about it. Without such a design, observed gradient differences cannot be attributed to the architecture itself.

8 Ethics, Privacy, Fairness, and Governance

A longitudinal record of AI-generated competency estimates concerning minors, retained across more than a decade and potentially accessible to institutions, raises

governance questions that the architecture must address explicitly rather than defer to implementation. The positions below are proposed design constraints, not solved problems.

8.1 Consent, Minors, and Data Rights

A learner aged six cannot give meaningful consent to a record that will follow them into adulthood. The proposed framework therefore treats guardian consent as necessary but not sufficient. On reaching majority the learner would hold rights of review, redaction, and export over the accumulated record, and specified categories, including behavioural and qualitative observer entries, would expire automatically. Data minimisation and purpose limitation would be defined in statute rather than left to platform policy, and secondary use for purposes not specified at collection would require fresh consent.

8.2 Access Regimes: Disclosure, Recognition, and Consent

Access to the portfolio divides into regimes distinguished by who holds the record and on what basis, and conflating them produces incoherent governance. The portfolio is generated within the state education system, so the state is already the custodian of the data. Access by non-state bodies rests on an entirely different footing. Employers and other private parties hold no claim on the record, so access is strictly consent-based, revocable, limited to the fields the learner selects, logged and visible to the learner, and structured so that refusal leaves no recorded trace. Universities occupy an intermediate position, since public institutions make selection decisions of their own, and are treated here under the consent regime.

Military intake is treated as voluntary disclosure rather than statutory access. This choice is deliberate. The portfolio supplements existing psychometric and medical classification, so a conscript who does not disclose is still assessed by the established process. Refusal is therefore substantively, not merely formally, available. Voluntary disclosure is also the least restrictive means of achieving the placement gains the architecture claims, which is the appropriate test where an existing legal obligation already constrains the individual in other respects.

Disclosure is framed here as recognition of contribution rather than as a neutral administrative option. This follows the two-directional engagement principle that governs the framework as a whole. The system invests in the learner in proportion to the learner's investment in it, and the point of intake is where that principle becomes consequential. A learner who has accumulated a substantial record and elects to make it available is offering the institution something of value, and the framework treats placement advantage as the appropriate return. The design intent is therefore that disclosure becomes the norm, and the consequence that non-disclosure grows informative as participation rises is accepted rather than mitigated.

Two constraints follow from that position. The first concerns interpretation: a thin portfolio may indicate low contribution, but it may equally indicate late platform deployment in a municipality, transfer between systems, or interrupted attendance. Distinguishing non-contribution from non-recording is a precondition for treating the record as evidence of contribution at all, and coverage metadata must accompany the record for this reason. The second concerns scope, and binds irrespective of disclosure rates: intake access is limited to defined competency fields, and childhood-era qualitative observer entries, recorded by adults about a child who never reviewed them,

are excluded. Where a state body other than the intake process seeks access, educational records applied to a different administrative purpose constitute a secondary use across agencies requiring explicit statutory authorisation rather than being implied by state custody.

Pre-conscription access forms a further case. Talent identification from age twelve concerns a person under no military obligation, so guardian consent is the only available basis, with the pressures discussed in Section 8.4.

8.3 Algorithmic Fairness and Contestability

A competency estimate produced by a probabilistic model may perform unevenly across language of instruction, community, and socioeconomic background. The framework would require bias auditing across these strata, testing of measurement invariance before any high-stakes use, and publication of audit results. These requirements are consistent with concerns raised in the wider literature on large language models in education, where questions of transparency, equity of access, and accountability remain unresolved [15]. Learners and guardians would hold a right to explanation of any consequential estimate and a route to human review, with teacher-mentor override recorded as such in the portfolio. An estimate that cannot be contested should not carry institutional weight.

8.4 Early Identification and Sorting Risk

Structured identification of aptitude from age twelve carries a risk that the architecture is intended to reduce but could equally amplify: early flagging may function as premature sorting, shaping expectations of learners and teachers alike. Proposed safeguards include a prohibition on the use of identification data for adverse or exclusionary decisions, and no effect on the learner pathway where a programme is declined. Periodic re-evaluation would keep early estimates from becoming self-confirming. Whether structured identification widens access more than it entrenches advantage is an empirical question for pilot evaluation.

9 Conclusion

This paper has proposed a cognitive infocommunications architecture for education. It integrates four components: competency-based rather than age-based grouping; the protege effect and Vygotsky's zone of proximal development as the cognitive basis for a mixed-age peer apprenticeship model; Bloom's mastery paradigm applied through an adaptive AI content delivery and benchmarking platform; and a longitudinal digital portfolio as the data infrastructure integrating system outputs into a continuously updated competency record. Support is hierarchical: the platform delivers instruction, peers remediate where it fails, and expert remediation is reserved for learners the peer layer cannot serve.

Situated within the CogInfoCom framework of Baranyi, Csapo, and Sallai [3], the proposal treats the learning environment as a cognitive infocommunication system in which the co-evolution of human cognitive capability and ICT architecture is structurally designed rather than incidental. The contribution is conceptual. The paper maps an architecture and its components, and does not demonstrate educational effectiveness, scalability, or institutional feasibility.

The architecture generates a set of testable hypotheses that formal specification and pilot evaluation must now address. They concern competency trajectory gains under mastery-gated progression, cognitive effects of structured near-peer mentoring, and the validity of longitudinal portfolio assessment relative to single-point examination. Future work will focus on specification of the competency state model, the privacy-preserving portfolio data architecture, and the design of a pilot study capable of testing these hypotheses.

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